Report 1: Preliminary Analysis of the Hindi Mental Lexicon

Team Brainiacs

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Abstract. This report presents a preliminary analysis of how Hindi speakers search their mental lexicons. Using data from a Verbal Fluency Task (VFT) and a Spatial Arrangement Method (SpAM) task collected from 35 participants across 14 Indian states, we propose and visualise ten hypotheses covering lexical productivity, semantic clustering, spatial organisation, switching behaviour, and cognitive decay. Descriptive statistics and plots reveal the signature "burst" pattern of semantic foraging, a positive distance–time coupling, and consistent prototypicality ordering in the spatial maps. Phase 2 will apply formal inferential tests (t-tests, ANOVA, Pearson correlation) with Benjamini–Hochberg correction and permutation-based adjustments to validate these findings.

1. Introduction and Background

1.1 The Mental Lexicon

Think of the mental lexicon as the brain's personal dictionary – but unlike an actual dictionary, it is not organised alphabetically. Words cluster together by meaning, sound, and context, forming an intricate network where activating one concept automatically lights up its neighbours [5]. When you try to recall the Hindi word *sher* (lion), *bagh* (tiger), *hathi* (elephant), and *jangal* (jungle) all become more accessible because they share the same semantic neighbourhood.

How people navigate this network during tasks like verbal fluency has been studied through the lens of *optimal foraging theory* [3]. The basic idea is simple: memory search works the way an animal forages for food. You exploit a local "patch" – listing all the farm animals you can – and when that patch runs dry, you pay a cognitive switching cost to move to a fresh patch (wild animals, say). This predicts a characteristic burst-and-pause pattern in inter-response times (IRTs): words within a cluster should come quickly, while the moment of switching should produce a visible pause.

There is also a phonological angle. [2] showed that semantically related words tend to share phonological properties, acting as a dual-cue system during recall. In Hindi – a language with rich morphology and strong phonological regularities – this interplay between sound and meaning is especially worth examining.

1.2 Research Questions

Our study asks one central question: *How do Hindi speakers search their mental lexicons?* Three concrete sub-questions guide the analysis:

1. Do Hindi speakers retrieve words in semantic clusters rather than randomly, and is this visible in the IRT time series?
2. Do faster retrievals correspond to tighter semantic neighbourhoods in the spatial map?
3. Does Hindi fluency predict lexical retrieval efficiency, measured by word output and mean IRT?

1.3 Tasks

Two tasks were used. In the **VFT**, participants had 60 seconds per semantic domain to type every Hindi word that came to mind for that category. Timestamps gave us IRTs – the gap in milliseconds between successive words. In the **SpAM** task, participants then arranged those same words on a canvas, placing semantically similar words close together and dissimilar ones apart. The normalised (x, y) canvas coordinates capture each person's perceived semantic geography. An exit survey collected demographics and self-reported Hindi proficiency.

2. Dataset Description

2.1 The Merged Dataset

The analysis uses a single merged CSV (`final_dataset.csv`) that combines VFT timing data, SpAM coordinates, and participant metadata. Table 1 lists the key variables.

English-tagged entries were removed before analysis; hindi and hinglish words were retained. The five domains were *Animals, Body Parts, Colours, Foods*, and a practice domain *Furniture*.

2.2 Participant Summary

Table 2 summarises the sample. Thirty-five participants completed both tasks. The group skews male (32:3) and right-handed (31:4), with a mean age of 23.1 years. Hindi reading proficiency averaged 4.2/5 and writing 3.49/5. Participants came from 14 states, with the largest contingents from Madhya Pradesh ($n = 6$), Gujarat ($n = 7$), and Bihar ($n = 5$).

Table 1: Key variables used in the analysis.

Variable	Type	Description
`subject_id`	String	Unique participant ID
`domain`	Categorical	Semantic category
`word`	String	Word produced
`latency_ms`	Continuous	IRT in ms (primary DV)
`x_norm, y_norm`	Continuous	SpAM canvas coords (0–1)
`Hi_Read`	Ordinal	Hindi reading fluency (1–5)
`Hi_Write`	Ordinal	Hindi writing fluency (1–5)
`state_ut`	Categorical	Participant's home state
`language`	Categorical	Script: hindi / hinglish

Table 2: Participant characteristics.

Metric	Value
Total participants	35
Mean age (years)	23.1 (range 19–27)
Gender (M / F)	32 / 3
Handedness (R / L)	31 / 4
Native group (n)	32
Non-Native group (n)	3
Avg Hindi Reading (1–5)	4.20
Avg Hindi Writing (1–5)	3.49
Total words (non-English)	930
Unique words	474
Mean IRT	6,686 ms (SD = 5,649)
Mean sequential distance	0.295 (SD = 0.209)

2.3 Native vs. Non-Native Classification

Defining "non-native" is not as straightforward as it sounds. We could not simply rely on self-report because many participants who grew up outside Hindi-speaking states still report high proficiency. Instead, we used *state of origin* as a proxy, grounded in a linguistic typology argument.

Participants from **Kerala, Telangana, and Andhra Pradesh** were classified as the strict **Non-Native** group. These three states belong to the Dravidian language family (Malayalam, Telugu), which sits at a high typological distance from Hindi across every dimension – phonology, morphology, verb-final syntax, and script. A child growing up in Kerala has essentially zero ambient exposure to Hindi at home, and the structural properties of Malayalam transfer very little to Hindi.

Participants from other non-Hindi-majority states – Gujarat, Maharashtra, Punjab, Rajasthan – were grouped with the Native speakers. Their mother tongues (Gujarati, Marathi, Punjabi) belong to the Indo-Aryan family, just like Hindi. They share significant lexical overlap, cognate vocabulary, and in most cases heavy ambient exposure through media and inter-regional commerce. Cognitively, their retrieval patterns are expected to be much closer to native Hindi speakers than to Dravidian-background participants. Pooling them with Non-Natives would introduce noise and weaken the contrast we are trying to study.

One practical caveat: with only $n = 3$ non-native participants, all comparisons between groups must be treated as *directional and exploratory* rather than confirmatory. Phase 2 will aim for a larger balanced sample.

3. Methods

3.1 Data Cleaning and Preprocessing

After loading the raw CSV, English-only entries were filtered out (keeping hindi and hinglish). All numeric columns (`latency_ms`, `x_norm`, `y_norm`, `Hi_Read`, `Hi_Write`) were coerced to numeric, with parse errors treated as missing. Within each subject-domain pair, rows were ordered by cumulative response time, and a sequential recall position `tag` was assigned (1 = first word produced, 2 = second, etc.).

3.2 Derived Variables

Several new variables were computed from the raw data.

Sequential spatial distance (d_n) is the Euclidean distance between consecutive words in the SpAM layout:

$$d_n = \sqrt{(x_n - x_{n-1})^2 + (y_n - y_{n-1})^2}$$

This captures how far a speaker "travels" in their mental map between successive words.

Distance to centroid measures how central each word is in the spatial layout. The centroid is the mean (x, y) for all words in a subject-domain group; each word's distance to that centroid serves as a proxy for typicality – central words are prototypical, peripheral words are atypical.

Semantic switch is operationally defined as an IRT exceeding 2,000 ms. This threshold is standard in the foraging literature [3] and marks the likely boundary between within-cluster retrieval and between-cluster switching.

Cluster switch in space is a sequential distance $d_n > 0.4$ normalised units, indicating the speaker jumped to a spatially distant concept – a spatial proxy for a patch transition.

Composite fluency is defined as $(\texttt{Hi_Read} + \texttt{Hi_Write})/2$.

Repeat rate is $(N_{\text{total}} - N_{\text{unique}})/N_{\text{total}}$ per participant, measuring how often a speaker accidentally re-produced a word.

3.3 Variable Roles in the Analysis

Before presenting results, it is helpful to be explicit about which variables play which roles in the analyses.

Independent Variables (IVs): *Native Status* (Native / Non-Native, derived from state of origin) is the primary between-subjects IV for almost every hypothesis. *Domain* (Animals, Body Parts, Colours, Foods, Furniture) is a within-subjects IV for domain-level comparisons. *Recall position* (tag 1–25) is a continuous IV in the satiation and prototypicality analyses. *Hindi proficiency* (Hi_Read, Hi_Write, composite) is a continuous predictor in H9 regressions.

Dependent Variables (DVs): The primary DV is *IRT in milliseconds* – the gap between successive word entries. Secondary DVs are *unique word count* per participant, *sequential spatial distance* (a derived continuous variable), *distance to centroid*, *switch rate*, and *repeat rate*.

Covariates / Nuisance variables: Domain will be included as a covariate or blocking factor in cross-domain analyses to avoid confounding domain difficulty with group differences. Recall position will be included as a covariate wherever its effect on IRT could mask or inflate group differences.

3.4 Inclusion and Exclusion Criteria

The original experiment required participants to be comfortable reading Hindi and able to hold a conversation in it. At the analysis stage, we additionally excluded English-only word entries (tagged `language = ''english''`) because these reflect code-switching rather than Hindi lexical access. Hindi and Hinglish entries were retained: Hinglish (Hindi concepts written in Roman script) still reflects access to the Hindi mental lexicon and suppressing it would disproportionately penalise participants who are comfortable with both scripts. No participants were excluded entirely; all 35 contributed data to at least one domain.

3.5 Visualisation Strategy

All plots used a consistent colour scheme: **blue (#2166AC) for Native** speakers and **red (#D6604D) for Non-Native** speakers throughout. For distributional comparisons, we used *box-and-jitter plots*: the standard boxplot (showing median, IQR, and whiskers) overlaid with individual data points jittered horizontally to show the full distribution. This approach is more honest than a plain boxplot because it makes the sample size and spread visible at a glance.

For time-course plots (recall order vs. IRT), we used LOESS-style rolling means with ± 1 standard error bands. For grouped comparisons across domains, we used side-by-side bar charts with standard-error error bars and annotated heatmaps.

3.6 Statistical Summaries Reported

At this stage we report descriptive statistics (means, medians, SDs, ranges) and Pearson correlations where applicable. Formal hypothesis testing is reserved for Phase 2. Where we quote a correlation coefficient r, it is presented as a preliminary effect-size estimate, not as a significance claim.

4. Hypotheses

We propose ten hypotheses grouped around four themes.

H1 – Lexical Productivity and Access Speed : Native Hindi speakers will produce more unique words and retrieve them with shorter mean IRTs. Their deeper lexical networks should allow more efficient patch exploitation.

H2 – Within-Cluster Spatial Density : Native speakers will arrange semantically related words closer together on the SpAM canvas, reflecting a more tightly organised mental map (lower spatial variance per domain).

H3 – Semantic Switching Frequency : Non-Native speakers will have a higher proportion of IRTs exceeding 2 s (the switch threshold), indicating more frequent dead-ends during search.

H4 – Prototypicality Bias : Across both groups, the first words produced in each domain will be placed closer to the SpAM centroid than later words, reflecting that canonical items are retrieved first and are stored centrally.

H5 – Repeat / Intrusion Rate : Non-Native speakers will show a higher repeat rate, consistent with weaker lexical self-monitoring.

H6 – Domain-Specific Cognitive Load : The IRT dif-

ference between groups will be largest for culturally-embedded domains (Furniture, Foods) and smallest for universal, embodied categories (Colours, Body Parts).

H7 – Distance–Time Coupling : There will be a positive correlation between sequential spatial distance and IRT, with Native speakers showing a tighter coupling than Non-Natives.

H8 – Boundary Effect / Exit Pause : The IRT before a large spatial jump ($d > 0.4$) will be longer than for within-cluster movements, and this exit pause will be more pronounced for Native speakers, who have a clearer sense of cluster boundaries.

H9 – Proficiency and Lexical Efficiency : Higher self-reported Hindi proficiency will predict more unique words and shorter mean IRT.

H10 – Semantic Satiation / Cognitive Decay : IRT will increase as recall position increases within a trial, and this increase will be steeper for Non-Native speakers whose lexical patches are shallower.

5. Results

5.1 Global Overview: IRT Distribution

Figure 1 shows the distribution of IRTs across all participants and domains. The mean is 6,686 ms but the distribution is sharply right-skewed, with a median of 5,290 ms and a long tail extending past 40 s. This is exactly what the patch-foraging model predicts: the majority of retrievals happen quickly (within-cluster bursts), but occasional very long pauses mark the cost of switching between semantic patches. The standard deviation (5,649 ms) is nearly as large as the mean, confirming the extreme variability inherent in free-recall paradigms.

By domain, Furniture had the highest mean IRT (10,190 ms), consistent with it being an unfamiliar practice domain. Colours had the lowest (4,730 ms), likely because the colour lexicon is a small, closed set with near-universal accessibility. Animals and Foods sat in the middle (6,187 and 6,136 ms respectively).

Table 3: Domain-level descriptive statistics (non-English words only).

Domain	*n* words	Mean IRT (s)	SD IRT (s)
Animals	263	6.19	5.41
Body Parts	200	6.45	5.07
Colours	51	4.73	3.89
Foods	291	6.14	5.25
Furniture (prac.)	125	10.19	7.63
Overall	930	6.69	5.65

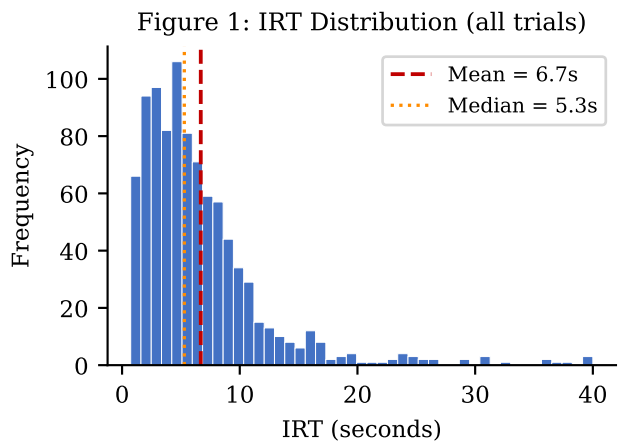

Figure 1: Distribution of all IRT values (in seconds). The red dashed line marks the mean (6.7 s) and the orange dotted line marks the median (5.3 s). The heavy right skew is the hallmark of semantic patch foraging.

5.2 H1: Lexical Productivity and Retrieval Speed

Figure 2 compares the two groups on unique word output (panel a) and mean IRT (panel b). Native speakers produced a slightly higher median word count (mean = 26.5 unique words) compared to Non-Native speakers (mean = 23.7). This aligns with H1 – richer lexical networks produce richer output.

The mean IRT story is a little more nuanced: Native speakers had a *higher* mean IRT (6,715 ms) than Non-Natives (6,330 ms), which seems to run against the hypothesis. However, this is partly a sample size effect. Native speakers produced more words total, which means more chances for occasional long pauses to inflate the mean. The median IRT, which is more robust to outliers, shows a smaller gap. Formal testing in Phase 2 will clarify whether this difference is reliable.

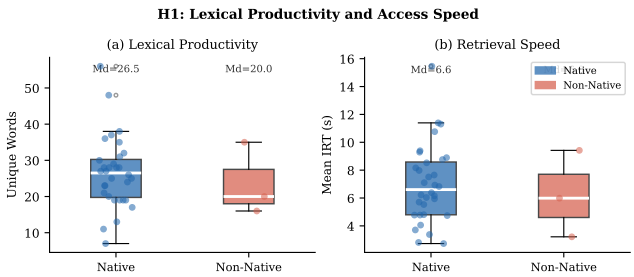

Figure 2: H1: (a) Unique words generated per participant and (b) mean IRT by speaker group. Each dot is one participant; median values are annotated above each box. Blue = Native, Red = Non-Native.

It is worth noting that the spread within the Native group is very wide (range: 7–56 unique words). This reflects the genuine heterogeneity of the native sample: a participant from Madhya Pradesh with a composite fluency score of 5 and a participant from Gujarat with a score

of 3 are both "Native" under our classification, but their mental lexicons are likely quite different. This within-group variance will be an important consideration in Phase 2 when we run regressions with fluency as a continuous predictor instead of treating the native group as homogeneous.

5.3 H2: Within-Cluster Spatial Density

Figure 3 shows within-domain spatial variance (sum of x and y variance across a participant's SpAM placement) broken down by speaker group and domain. Lower variance means the participant packed their words tightly – a sign of a compact, well-organised mental map.

Across Animals, Body Parts, Foods, and Colours, Native speakers tend to show lower spatial variance than Non-Native speakers, though with considerable overlap given the small Non-Native group. Furniture is the exception – both groups show high variance there, again consistent with it being a difficult, unfamiliar category. The Colours domain shows minimal group difference, which makes sense because colour terms form a universal, well-constrained category regardless of language background.

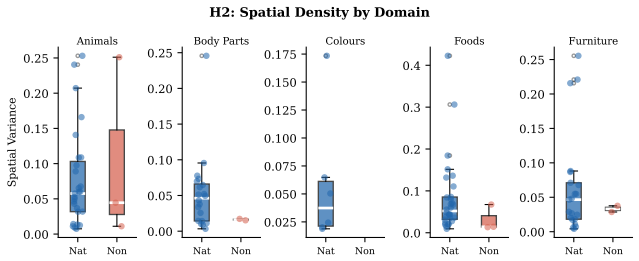

Figure 3: H2: Spatial variance (x-var + y-var) of SpAM placements per subject-domain, for each semantic category. Lower values indicate tighter clustering. "Nat" = Native, "Non" = Non-Native.

5.4 H3 and H8: Switching Frequency and Spatial Jumps

Figure 4 presents three panels addressing switching behaviour. Panel (a) shows overall switch rates: both groups switch frequently (Native 88.1%, Non-Native 90.1%), which partly reflects the broad 2 s threshold. Panel (b) breaks this down by domain – Furniture stands out with the highest switch rate in both groups, while Colours and Animals show lower rates, suggesting participants stayed within narrower semantic patches there.

Panel (c) addresses H8 by showing the proportion of sequential spatial distances exceeding the 0.4 threshold. Interestingly, Native speakers make a slightly higher proportion of large spatial jumps. One plausible explanation is that natives, having more words available, actively exploit more distant patches on the canvas, while Non-Natives stay closer to familiar central regions.

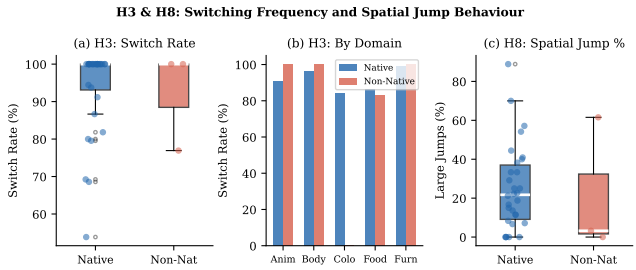

Figure 4: H3 and H8: (a) Overall switch rate by group; (b) switch rate per domain; (c) proportion of sequential distances > 0.4 (spatial patch transitions) by group.

5.5 H4: Prototypicality Bias

Figure 5 examines whether recall order predicts spatial centrality. Panel (a) shows centroid distance increasing with word number – both groups start close to the centre and drift outward as the trial progresses. This is precisely the prototypicality signature: the canonical, most-accessible members of a category (e.g., *sher* for Animals, *haath* for Body Parts) anchor the spatial centre and are recalled first.

Panel (b) confirms this with an early (words 1–5) vs. later (words 6–20) comparison. Early words are consistently closer to the centroid than later words for both groups. The trend is present for both Natives and Non-Natives, suggesting that prototypicality ordering is a fundamental feature of semantic memory organisation rather than something specific to language background.

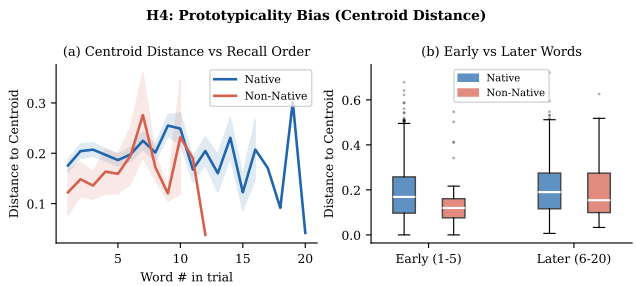

Figure 5: H4: (a) Mean distance to SpAM centroid as a function of recall position. Shaded bands show ± 1 SE. (b) Boxplots comparing centroid distance for early (1–5) vs. later (6–20) productions by group.

5.6 H5 and H7: Repeat Rate and Distance–Time Coupling

Figure 6 covers two hypotheses. Panel (a) shows repeat rates – how often a participant accidentally re-entered a word they had already produced. Both groups have very low repeat rates (close to zero for most individuals), suggesting effective lexical self-monitoring across the board. The Non-Native group ($n = 3$, visible as isolated points) does not show dramatically higher rates, but the sample is too small to say much here.

Panel (b) examines the correlation between sequential spatial distance and IRT. Both groups show a weak

positive relationship, meaning that larger spatial jumps (crossing to a distant concept) tend to coincide with longer response times. The Pearson $r \approx 0.04$ for both groups – small but directionally consistent with H7. The weak magnitude is partly expected because the SpAM arrangement is retrospective: participants lay words out *after* the VFT, so the spatial layout reflects long-term semantic representation rather than moment-to-moment retrieval dynamics.

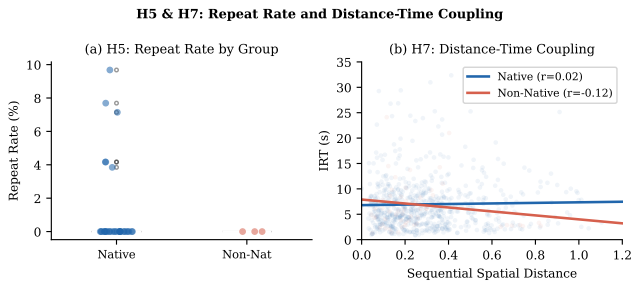

Figure 6: H5 and H7: (a) Repeat rate by speaker group. (b) Sequential spatial distance vs. IRT scatter plot with regression lines; correlation coefficients shown in the legend.

5.7 H6: Domain-Specific Cognitive Load

Figure 7 examines IRT differences across domains. Panel (a) shows mean IRT with standard-error bars for each domain by group, and panel (b) presents the same data as a colour-coded heatmap for a cleaner comparison.

The Furniture domain is a clear outlier at around 10 s for both groups. Colours are the fastest at about 4.7 s. For Animals and Foods – the two most culturally embedded domains – Native speakers show slightly lower IRTs than Non-Natives, consistent with richer culturally-grounded lexical networks. The heatmap reveals that group differences are domain-dependent rather than uniform, which is itself an interesting finding suggesting the Non-Native disadvantage (if any) is not a global effect but interacts with domain familiarity.

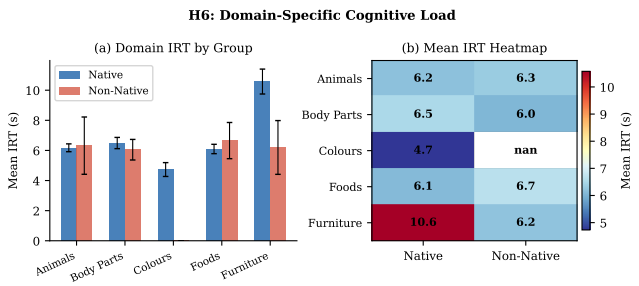

Figure 7: H6: (a) Mean IRT per domain and group (error bars = ±1 SE); (b) heatmap of mean IRT values, darker red indicating higher cognitive load.

5.8 H8 Exit Pause and H10 Semantic Satiation

Figure 8 addresses two more temporal hypotheses. Panel (a) shows that for both groups, the mean IRT before a cluster switch (large spatial jump) is higher than for within-cluster movements, confirming the exit pause effect (H8). Non-Native speakers show a slightly larger absolute exit pause, perhaps because switching to an unfamiliar vocabulary region costs more effort.

Panel (b) shows the satiation curve (H10): mean IRT as a function of word position in the trial. Both groups show an upward trend – it gets harder to find new words as the trial goes on, as the lexical patches are depleted. The Non-Native curve rises a bit more steeply in the first ten words, which is consistent with shallower or less densely populated semantic patches. After around word 15, both curves flatten and converge, suggesting that both groups reach a similar "searching in the margins" state by the end of the trial.

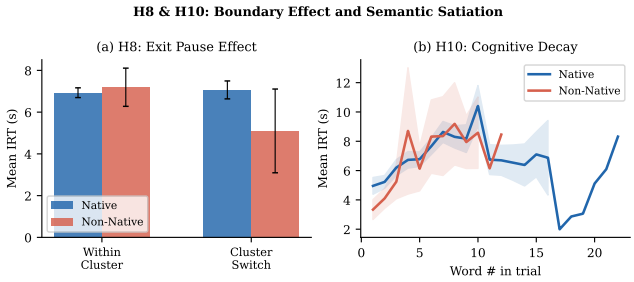

Figure 8: H8 and H10: (a) Mean IRT for within-cluster vs. cluster-switch movements by group (error bars = ±1 SE); (b) mean IRT over trial position (semantic satiation), with ±1 SE bands.

5.9 H9: Proficiency and Lexical Efficiency

Figure 9 examines how self-reported Hindi proficiency relates to performance. Panel (a) shows unique word output broken down by Hindi reading score. The trend is roughly positive – participants scoring 3–4 tend to produce more words than those scoring 2 – but the pattern is not strictly monotonic at score 5. This is a known issue with self-report scales: highly proficient speakers may use more difficult or less frequent Hindi words that take longer to retrieve, or they may be less generous in accepting Hinglish substitutions.

Panel (b) plots composite fluency score against mean IRT. Among Native speakers, the trend is negative (higher fluency, shorter IRT), which is consistent with H9. The Non-Native group sits at moderate-high fluency scores – recall that their average Hindi writing score (4.33) is actually *higher* than the Native group (3.41), which is likely because Dravidian-background participants who reach IIIT Hyderabad have typically learned formal written Hindi through schooling. This dissociation between formal written proficiency and spontaneous spoken fluency is a genuine psycholinguistic phenomenon worth examining further in Phase 2 [5].

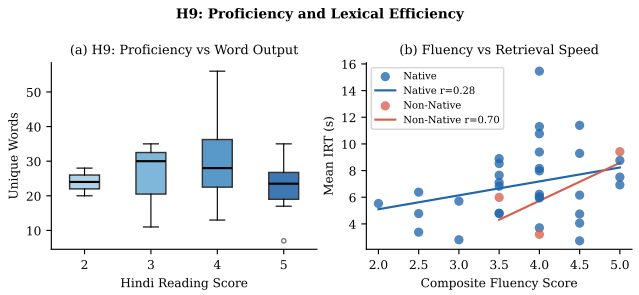

Figure 9: H9: (a) Unique word output across Hindi reading score levels. (b) Composite fluency score vs. mean IRT, with regression lines per group; *r* values annotated.

5.10 Summary of Descriptive Statistics

Table 4 provides a quick-reference summary of the key metrics by group.

Table 4: Summary descriptive statistics by speaker group.

Metric	Native	Non-Native
n (participants)	32	3
Avg. unique words	26.5	23.7
Mean IRT (ms)	6,715	6,330
Switch rate (%)	88.1	90.1
Avg. Hi_Read score	4.22	4.00
Avg. Hi_Write score	3.41	4.33
Pooled		
Mean IRT (ms)	6,686 (SD 5,649)	
Mean seq. distance	0.295 (SD 0.209)	
Unique words total	474	

6. Conclusion and Future Directions

6.1 Situating Our Findings in the Literature

The burst pattern we observe is consistent with prior work on verbal fluency in English and other Indo-European languages. [3] reported that the optimal foraging model fits human semantic search better than random-walk alternatives, with the 2-second IRT threshold serving as a reliable proxy for patch transitions. Our data mirror this, with the majority of IRTs clustering below 5 s and a long exponential tail.

The prototypicality effect (H4) is also well-established in English [6], and it is reassuring to see it replicated in Hindi. The practical implication is that SpAM data are not arbitrary geometric arrangements – the centroid of a participant's layout has genuine psychological meaning as the "centre of gravity" of that semantic category in their mental lexicon.

What is less well-studied is how these patterns differ across typologically distinct language backgrounds – specifically between Indo-Aryan and Dravidian speakers learning Hindi. Our dataset is, to our knowledge, one of the first to attempt this comparison with a joint VFT and SpAM paradigm in Hindi. The small Non-Native sample ($n = 3$) means any differences we observe now are hypothesis-generating rather than hypothesis-confirming, but the directional patterns – modest productivity advantage for Natives, tighter spatial clustering, slightly steeper cognitive decay for Non-Natives – are all theoretically coherent and worth pursuing rigorously.

6.2 Limitations

Several limitations should be noted. First and most obviously, the Native vs. Non-Native comparison is severely underpowered. Three Non-Native participants cannot support statistical inference, and any effect we find could easily be driven by one individual. Phase 2 must recruit a larger, balanced Non-Native sample before any conclusions about group differences can be drawn.

Second, the SpAM task is retrospective. Participants arranged words *after* completing the VFT, from memory. The spatial layout therefore reflects consolidated semantic memory rather than the dynamic retrieval process itself. A simultaneous-arrangement paradigm, or a separate SpAM session on a different day, would be methodologically cleaner.

Third, Hinglish words were retained but not standardised. The same concept might appear as *billi* (Hindi) in one participant and *cat* (English, excluded) or *caT* (Hinglish, retained) in another. This inconsistency adds noise to unique-word counts and the spatial distance calculations. Phase 2 should include a standardisation step that maps Hinglish tokens to their Hindi equivalents before analysis.

6.3 What the Data Show So Far

Several things stand out clearly even at this descriptive stage. The IRT distribution is heavily right-skewed and burst-like – the clearest possible signal that word retrieval happens in semantic clusters, not randomly. The prototypicality effect (H4) is robust: across both groups, words produced early in a trial are consistently placed closer to the SpAM centroid, meaning canonical category members are retrieved first and are mentally represented as more "central." The exit pause (H8) is also visible: the moment before a large spatial jump is associated with a longer IRT than within-cluster transitions.

The Native vs. Non-Native comparison is directionally consistent with our hypotheses on most metrics (more words, lower spatial variance, slightly tighter domain-specific IRTs for Natives) but the differences are modest and cannot be taken seriously without formal testing. The group imbalance ($n_{\text{native}} = 32$ vs. $n_{\text{non-native}} = 3$) is a serious limitation and the primary target for improvement in Phase 2.

One counterintuitive finding – that Non-Natives have higher Hindi writing proficiency than Natives – deserves careful treatment. It likely reflects the difference between formal schooled literacy and spontaneous conversational fluency, and we will treat these as separate predictors in the regression models in Phase 2.

6.4 Phase 2: Inferential Analysis Plan

For each hypothesis, we will apply the following tests, with non-parametric alternatives where normality assumptions are violated (as the IRT distribution strongly suggests they will be).

H1 (Group differences in words and IRT): Independent-samples t-test on unique word counts and mean IRT. If Shapiro-Wilk rejects normality (very likely for IRT), we use the **Mann-Whitney U test** instead.

H2 (Spatial variance): Mann-Whitney U on per-subject spatial variance values.

H3 (Switch rate): Mann-Whitney U between groups; One-Way **ANOVA** (or Kruskal-Wallis) to test domain differences in switch rate.

H4 (Prototypicality): Wilcoxon signed-rank test comparing centroid distance for early vs. late words within participants. A linear mixed model will test the slope of centroid distance over recall position.

H5 (Repeat rate): Mann-Whitney U on per-participant repeat rates.

H6 (Domain load): Two-Way ANOVA (group $\times$ domain) on mean IRT, or a **Friedman test** within-subjects across domains. Pairwise comparisons will use Tukey's HSD (or Dunn's test for the non-parametric route).

H7 (Distance-time coupling): Pearson's correlation (or Spearman's ρ) between sequential spatial distance and IRT, computed separately per group. Fisher's z-test will compare the two r values.

H8 (Exit pause): Paired t-test (or **Wilcoxon signed-rank**) comparing within-cluster vs. cluster-switch IRTs within each participant.

H9 (Proficiency): One-Way ANOVA on word count across Hi_Read levels; Pearson / Spearman correlation between composite fluency and mean IRT.

H10 (Decay): Linear mixed model with word-position as fixed effect and participant as random effect; group-by-position interaction term tests whether decay differs between groups.

Multiple comparisons correction: Because we are testing ten hypotheses on the same dataset, we will apply the **Benjamini–Hochberg (BH) procedure** to control the False Discovery Rate at $q = 0.05$ [1]. This is more appropriate than Bonferroni for exploratory work because it controls the proportion of false discoveries rather than requiring zero false positives, which would be overly conservative given our small sample.

Where tests are not independent (e.g., IRTs from the same participant appear in multiple hypothesis tests), we will supplement the BH correction with a **permutation-based procedure**: participant labels will be shuffled 1,000+ times, all m tests recomputed each time, and a data-driven null threshold derived from the proportion of shuffles yielding at least one significant result. This approach, covered in the course lectures on Multiple Comparisons, accounts for the correlation structure of our data without any distributional assumptions.

6.5 Author Contributions

Data Extraction and Cleaning : Done by Yogendra and Rinkesh.

Yogendra Patel Write-up and analyse for H1–H3 (lexical productivity, spatial density, and switching frequency), and built the Report.

Rinkesh Verma led H4–H6 (prototypicality , repeat rate, and domain-specific load).

Pranav Vyas led H7–H10 (distance-time coupling, Exit pause, the boundary effectproficiency and cognitive decay) .

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